

CareConnect: Patient-Provider Electronic Health Record (EHR)

Comprehensive System Modules, Use Cases Specification & Technical Architecture

Document Version:	2.0.0 (Production Verified)	Target Platform:	Web (React 19 + Spring Boot 3)
Live Frontend:	https://careconnect-ehr.vercel.app/	Live Backend API:	https://careconnect-backend-uim9.onrender.com
GitHub Monorepo:	https://github.com/faizan0264/careconnect-ehr	Database DDL:	Oracle Database 19c/21c (CLOB/Cascades)

1. Executive Overview & System Purpose

CareConnect EHR is an enterprise-grade hospital information and clinical consultation portal designed to bridge the systemic disconnect between healthcare providers and patients. The system enforces strict **Role-Based Access Control (RBAC)** across Doctors, Patients, and Administrators, while eliminating clinical blindspots through real-time **Drug-Allergy Interaction Checks**, **Physician Schedule Collision Prevention**, a dual-access **Medical Report & Diagnostic Document Vault**, and an immutable **HIPAA Audit Trail Engine**.

2. System Modules Architecture Matrix

Module #	Module Name	Primary Actor	Key Functional Scope	Compliance / Standards
MOD-01	Authentication & Access Governance	All (Doctor, Patient, Admin)	Strict zero-phantom login, credential hashing, profile credential updates, session management.	NIST 800-63B / HIPAA Access Control
MOD-02	Doctor Clinical Workstation	Doctor / Physician	Clinic queue, longitudinal chart inspection, full SOAP notes (Subjective, Objective, Assessment, Plan), real-time vitals.	HL7 CDA / SOAP Standard
MOD-03	Computerized Physician Order Entry (CPOE)	Doctor / Physician	Electronic orders for Laboratory panels and Radiology scans with priority flags (Routine/STAT) and result fulfillment.	ONC CPOE Certification Criteria
MOD-04	e-Prescribing & Allergy Safety Engine	Doctor / Physician	Prescription dispatch with active allergy cross-referencing, critical contraindication modals, and formal override auditing.	FDA Adverse Drug Safety Guidelines
MOD-05	Patient Health Portal & AVS	Patient	Longitudinal health overview, After-Visit Summary (AVS) review with 1-click browser printing, pharmacy refill requests.	Meaningful Use Stage 3 Patient Access
MOD-06	Conflict-Free Appointment Scheduling	Patient / Doctor	Real-time physician schedule collision detection, busy-slot locking, appointment status updates, cancellation.	ACID Transaction Isolation
MOD-07	Medical Report & Diagnostic Vault	Doctor & Patient	Dual-access Base64 file upload (PDF/PNG/JPG up to 15MB), in-browser image preview, clinical findings, one-click file download.	DICOM / CLOB Document Storage

MOD-08	Admin Console & HIPAA Audit Trail	Administrator	User provisioning (add doctors/nurses), account deactivation, immutable HIPAA audit log capturing IP, user, action, and timestamp.	45 CFR § 164.312(b) Audit Controls
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3. Detailed Use Cases & Execution Workflows

UC-01: Physician Drug-Allergy Safety Interception & Override

- **Primary Actor:** Attending Physician (ROLE_DOCTOR)
- **Precondition:** Patient has documented drug allergies in chart (e.g., Penicillin for John Doe).
- **Trigger:** Physician prescribes contraindicated antibiotic (e.g., Amoxicillin 500mg).
- **Standard Flow:** 1. System detects therapeutic allergen match. 2. Halts order dispatch. 3. Displays Critical Allergy Alert modal. 4. Physician reviews risk, selects clinical justification ('Clinical benefit outweighs risk with close monitoring'), checks legal responsibility acknowledgment. 5. Order commits to e-Prescription log. 6. Audit engine writes DOCUMENT_ORDER event with override rationale.
- **Postcondition:** Safe clinical audit trail maintained; pharmacy alerted to monitored override.

UC-02: Patient Conflict-Free Appointment Scheduling

- **Primary Actor:** Patient (ROLE_PATIENT)
- **Precondition:** Patient is logged into Portal; physician clinic schedule exists.
- **Trigger:** Patient selects physician, target date, and requested consultation time.
- **Standard Flow:** 1. Interface queries physician appointments for target date. 2. Occupied slots are dynamically flagged with 'Busy ●' and disabled. 3. If patient attempts selecting an occupied slot, an active Conflict Warning alert appears. 4. Patient selects an open slot and submits. 5. Backend validates concurrency and persists appointment. 6. HIPAA audit log captures APPOINTMENT_SCHEDULED.
- **Postcondition:** Double-booking is completely prevented across frontend and API layers.

UC-03: Dual-Access Medical Report Vault & File Ingestion

- **Primary Actor:** Patient (ROLE_PATIENT) or Doctor (ROLE_DOCTOR)
- **Precondition:** User has a diagnostic document (PDF, PNG, JPG up to 15MB).
- **Trigger:** User clicks 'Upload Document' or 'Upload Scan / Report'.
- **Standard Flow:** 1. User drags and drops file or selects from filesystem. 2. HTML5 FileReader encodes file to Base64 CLOB. 3. System auto-categorizes document (Radiology, Lab, Pathology, etc.). 4. User adds clinical findings and date. 5. Document is stored in database with clear attribution tag ('Physician File' vs 'Patient Record'). 6. Both actors can inspect scan preview in-browser and download file.
- **Postcondition:** Document securely attached to chart; DOCUMENT_UPLOADED written to audit log.

UC-04: HIPAA Security Audit Trail Verification

- **Primary Actor:** Hospital Compliance Officer / Administrator (ROLE_ADMIN)
- **Precondition:** Administrator is authenticated to the Admin Console.
- **Trigger:** Administrator accesses HIPAA Audit Trail tab.

- **Standard Flow:** 1. System queries immutable AUDIT_LOGS table. 2. Renders chronologically ordered security events capturing user identity, assigned role, client IP address, action type, and exact timestamp. 3. Administrator filters by action (VIEW_RECORD, CREATE_ORDER, DOCUMENT_UPLOADED, DOCUMENT_DELETED).
- **Postcondition:** Full forensic traceability for regulatory compliance audits.

4. Oracle Database 19c/21c Enterprise Schema Specification

The relational schema is defined in **CareConnect_Oracle_Schema.sql**. It enforces strict referential integrity, check constraints, foreign keys with ON DELETE CASCADE, and performance B-Tree indexes.

Table Name	Primary Key	Foreign Keys	Key Columns & Data Types	Integrity Constraints
USERS	USER_ID (NUMBER)	None	USERNAME VARCHAR2(100), PASSWORD_HASH VARCHAR2(255), ROLE VARCHAR2(50), FULL_NAME VARCHAR2(150), EMAIL VARCHAR2(150)	UNIQUE(USERNAME), CHECK(ROLE IN ('ROLE_DO CTOR','ROLE_PATIENT','ROL E_ADMIN'))
PATIENTS	PATIENT_ID (NUMBER)	USER_ID -> USERS	MRN VARCHAR2(50), FIRST_NAME VARCHAR2(100), LAST_NAME VARCHAR2(100), DOB DATE, ALLERGIES VARCHAR2(500), BLOOD_GROUP VARCHAR2(10)	UNIQUE(MRN), ON DELETE CASCADE
APPOINTMENT S	APPT_ID (NUMBER)	PATIENT_ID -> PATIENTS, DOCTOR_ID -> USERS	APPT_DATE VARCHAR2(20), TIME_SLOT VARCHAR2(20), REASON VARCHAR2(500), STATUS VARCHAR2(50)	CHECK(STATUS IN ('Confirm ed','Cancelled','Completed')), B-Tree Index on (DOCTOR_ID, APPT_DATE)
ENCOUNTERS	ENCOUNTER_ID (NUMBER)	PATIENT_ID -> PATIENTS	DOCTOR_NAME VARCHAR2(150), VITALS_BP VARCHAR2(20), VITALS_HR NUMBER, SOAP_SUBJECTIVE CLOB, SOAP_PLAN CLOB	ON DELETE CASCADE
DIAGNOSTIC_O RDERS	ORDER_ID (NUMBER)	PATIENT_ID -> PATIENTS	ORDER_NAME VARCHAR2(255), ORDER_TYPE VARCHAR2(50), PRIORITY VARCHAR2(20), RESULT CLOB, STATUS VARCHAR2(50)	CHECK(PRIORITY IN ('Routine','Urgent','STAT'))
PRESCRIPTION S	RX_ID (NUMBER)	PATIENT_ID -> PATIENTS	DRUG_NAME VARCHAR2(255), DOSAGE VARCHAR2(100), FREQUENCY VARCHAR2(100), OVERRIDE_REASON VARCHAR2(500), STATUS VARCHAR2(50)	ON DELETE CASCADE
MEDICAL_REP ORTS	REPORT_ID (NUMBER)	PATIENT_ID -> PATIENTS	TITLE VARCHAR2(255), REPORT_TYPE VARCHAR2(50), FILE_NAME VARCHAR2(255), FILE_DATA CLOB, UPLOADED_BY VARCHAR2(150), UPLOADER_ROLE VARCHAR2(50)	CHECK(REPORT_TYPE IN ('L ABORATORY','RADIOLOGY',' PATHOLOGY','PRESCRIPTIO N','DISCHARGE_SUMMARY',' OTHER'))
AUDIT_LOGS	LOG_ID (NUMBER)	None (Independent)	USER_NAME VARCHAR2(150), ACTION VARCHAR2(100), DETAILS VARCHAR2(1000), IP_ADDRESS VARCHAR2(50), TIMESTAMP TIMESTAMP	B-Tree Index on (TIMESTAMP), Immutable Append-Only

5. Spring Boot REST API Endpoint Directory

Method	Endpoint URI	Controller	Payload / Parameters	Description & Audit Action
POST	/api/v1/auth/login	AuthController	{ username, password }	Authenticates credentials; returns user profile.
POST	/api/v1/auth/signup	AuthController	{ fullName, email, password, dob... }	Registers new patient; generates unique MRN.
PUT	/api/v1/auth/profile	AuthController	{ currentUsername, newUsername, newPassword }	Updates credentials; logs security audit event.

GET	/api/v1/patients	PatientController	None	Retrieves all clinical patient records.
POST	/api/v1/appointments/book	AppointmentController	{ patientId, doctorId, date, timeSlot }	Enforces collision check; logs APPOINTMENT_SCHEDULED.
POST	/api/v1/reports/upload	MedicalReportController	{ patientId, title, fileData, notes... }	Stores Base64 scan; logs DOCUMENT_UPLOADED.
GET	/api/v1/reports/patient/{id}	MedicalReportController	patientId	Retrieves all diagnostic reports for chart.
DELETE	/api/v1/reports/{id}	MedicalReportController	id, deletedBy	Removes document; logs DOCUMENT_DELETED.
GET	/api/v1/audit/logs	AuditController	action (optional)	Returns HIPAA immutable audit events.
GET	/swagger-ui.html	OpenAPI / Swagger	None (Interactive UI)	Interactive live API sandbox and documentation.